

SUJIT KUMAR SAHOO

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RESEARCH INTERESTS

Particle Physics and Cosmology: Dark matter Phenomenology in connection with Neutrino Physics and Post-inflationary Cosmology.

EDUCATION

PH.D. | 2021 (JULY) - 2026 (JUNE). | IIT HYDERABAD

- Affiliated to: IIT HYDERABAD
- Guide: Prof. Narendra Sahu
- Funding Agency: UGC CSIR, India
- Thesis Title: "Exploring Physics Beyond the Standard Model in light of Neutrino mass, Dark Matter and Effective Light Degrees of Freedom."

M.SC. (PHYSICS) | 2018-20 | IIT ISM DHANBAD

- Affiliated to: IIT ISM DHANBAD
- Electives: High Energy Physics, Plasma and Space Physics, Quantum Computation
- Grade: 8.34/10

B.SC. (PHYSICS) | 2015-18 | BHADRAK AUTONOMOUS COLLEGE, BHADRAK

- Affiliated to: Fakir Mohan University, Odisha
- Percentage Secured: 78.20%

12TH (SCIENCE) | 2013-15 | BHADRAK JR. COLLEGE, BHADRAK

- Board: CHSE, Odisha
- Percentage Secured: 78.7%

10TH | 2012-13 | KRUPASINDHU HIGH SCHOOL, ADALPANK, BHADRAK

- Board: BSE, Odisha
- Percentage Secured: 86.7%

NATIONAL LEVEL EXAMS QUALIFIED

- National Mean-cum-Merit Scholarship, 2010-11
- JAM 2018
- GATE 2020 & GATE 2021
- CSIR-UGC NET JRF 2020 JULY

PROJECTS

- B.Sc.: The uncertainty principle and Einstein's radiation box | Guide- Dr. Rajat Kumar Pradhan, Bhadrak Autonomous College
- M.Sc.: (1) A brief discussion on neutrino oscillation and CP violation | Guide- Prof. Sanjib Kumar Agarawalla, Institute of Physics, Bhubaneswar

(2) Synthesis and characterization of undoped & Fe doped SnS_2 nanoparticles for solar cell application | Guide- Prof. Shailendra Kumar Sharma, IIT ISM Dhanbad

SKILLS & ABILITIES

COMPUTER

- Operating Systems: Windows, Linux, Mac
- Software/Packages : LanHEP, CalcHEP, MicrOMEGAs, Cosmo Transition, Package-X, LoopTools, Veusz, ORIGIN, Mathematica, Python, Julia.

COMMUNICATION

- Odia (Mother tongue), English, Hindi

SCHOOLS/WORKSHOPS/CONFERENCES

- HSF-India Scientific Computing Workshop (2026), IIT Hyderabad.
- Workshop on High Energy Physics Phenomenology (WHEPP 2025), IIT Hyderabad (Talk Title- "Gravitational Echos of Singlet-Doublet DM Induced Radiative Neutrino Mass.")
- Indian Institutions-Fermilab Collaboration in Neutrino Physics (IIFC-vP) School (2025), NISER Bhubaneswar.
- PHOENIX-2025 at IIT Hyderabad (Talk Title- "Viability of post-inflationary freeze-in with precession cosmology.")
- XXVI DAE-BRNS High Energy Physics Symposium – 2024, Banaras Hindu University, Varanasi (Talk Title- "Effective Theory of Light Dirac Neutrino portal Dark Matter with Observable ΔN_{eff} ")
- Kashiwa-no-ha Dark Matter and Cosmology Symposium (2024), Kavli IPMU, Tokyo University (Poster Title- "Dirac Neutrinos via Lepton Quarticity and Self-Interacting Dark Matter")
- COSMO'24 at Yukawa Institute for Theoretical Physics, Kyoto University (Talk Title- " ν_R -philic dark matter and ΔN_{eff} through an EFT approach.")
- Workshop on Dark Matter and Astroparticle Physics (WDMAP 2024), IOP Bhubaneswar (Poster Title "An Effective Field Theory approach to ΔN_{eff} and Dark Matter in light of Dirac Neutrinos.")
- International Conference on Future Prospects in Neutrino and Astroparticle Physics (ICFPNAP 2024), Assam Don Bosco University (Talk Title- "An approach to Dirac neutrino mass with Singlet-Doublet dark matter and ΔN_{eff} ")
- PHOENIX-2023 at IIT Hyderabad (Talk Title- "Singlet-doublet fermionic dark matter with Dirac neutrino mass, $(g-2)_\mu$ and ΔN_{eff} ")
- ML4HEP School on Statistical Methods and Machine Learning in High Energy Physics (2023), ICTS Bangalore.
- ICFA School on Instrumentation in Elementary Particle Physics (ICFA-2023), TIFR Mumbai.
- Data and Machine Learning at The Large Hadron Collider (DML@LHC-2022), IIT Hyderabad.

PUBLICATIONS

- Inelastic Singlet-Doublet Fermion Dark Matter in light of the 248 keV LZ event. Debasish Borah, **SKS**, Narendra Sahu, and Shashwat Sharma. Arxiv: [2609.07800](https://arxiv.org/abs/2609.07800).
- Singlet-Doublet fermion origin of dark matter, neutrino mass and inverse first-order electroweak phase transition. Debasish Borah, Indrajit Saha, **SKS**, Narendra Sahu, and Shashwat Sharma. Arxiv: [2608.12483](https://arxiv.org/abs/2608.12483).

- Anatomy of RHN DM relic in the vanilla scotogenic neutrino mass model. **SKS**, Narendra Sahu, and Vicky Singh Thounaojam. Arxiv: [2601.00436](#).
- Gravitational Wave Probe of Singlet-Doublet Dark Matter Induced Radiative Neutrino Mass. Ujjal Kumar Day, Santu Kumar Manna, Partha Kumar Paul, **SKS**, and Narendra Sahu. Arxiv: [2511.19386](#). DOI: [JHEP 07 \(2026\) 264](#).
- Impact of conversion-driven processes on singlet-doublet Majorana dark matter relic. Partha Kumar Paul, **SKS**, and Narendra Sahu. Arxiv: [2511.14571](#). DOI: [Phys.Rev.D 113 \(2026\) 9, 095040](#).
- Viability of post-inflationary freeze-in with precision cosmology. Anirban Biswas, Sougata Ganguly, Dibyendu Nanda, **SKS**. Arxiv: [2505.13624](#).
- Effective theory of light Dirac neutrino portal dark matter with observable ΔN_{eff} . Debasish Borah, Satyabrata Mahapatra, Dibyendu Nanda, **SKS**, and Narendra Sahu. Arxiv: [2502.10318](#). DOI: [Phys.Rev.D 112 \(2025\) 5, 055010](#).
- Anatomy of singlet-doublet dark matter relic: annihilation, co-annihilation, co-scattering, and freeze-in. Partha Kumar Paul, **SKS**, and Narendra Sahu. Arxiv: [2412.02607](#). DOI: [JCAP 10 \(2025\) 053](#).
- Self Interacting Dark Matter and Dirac neutrinos via Lepton Quarticity. Satyabrata Mahapatra, **SKS**, Narendra Sahu, and Vicky Singh Thounaojam. Arxiv: [2312.12321](#). DOI: [Phys.Rev.D 109 \(2024\) 5, 055036](#).
- Singlet-doublet fermion dark matter with Dirac neutrino mass, $(g - 2)_\mu$ and ΔN_{eff} . Debasish Borah, Satyabrata Mahapatra, Dibyendu Nanda, **SKS**, and Narendra Sahu. Arxiv: [2310.03721](#). DOI: [JHEP 05 \(2024\) 096](#).

EXPERIENCE

TEACHING ASSISTANT | IIT HYDERABAD

- Courses: Mathematical Physics, Quantum Mechanics.

GUEST FACULTY | BHADRAK AUTONOMOUS COLLEGE, BHADRAK | FEB 2021-JULY2021

- Courses: Mathematical Physics (UG Physics Hons.), Mathematical Physics(PG), Electricity & Magnetism (UG Physics pass), Thermodynamics (UG Physics Pass).

I hereby declare that the above written particulars are true to best of my knowledge and belief.

August 2026

Sujit Kumar Sahoo